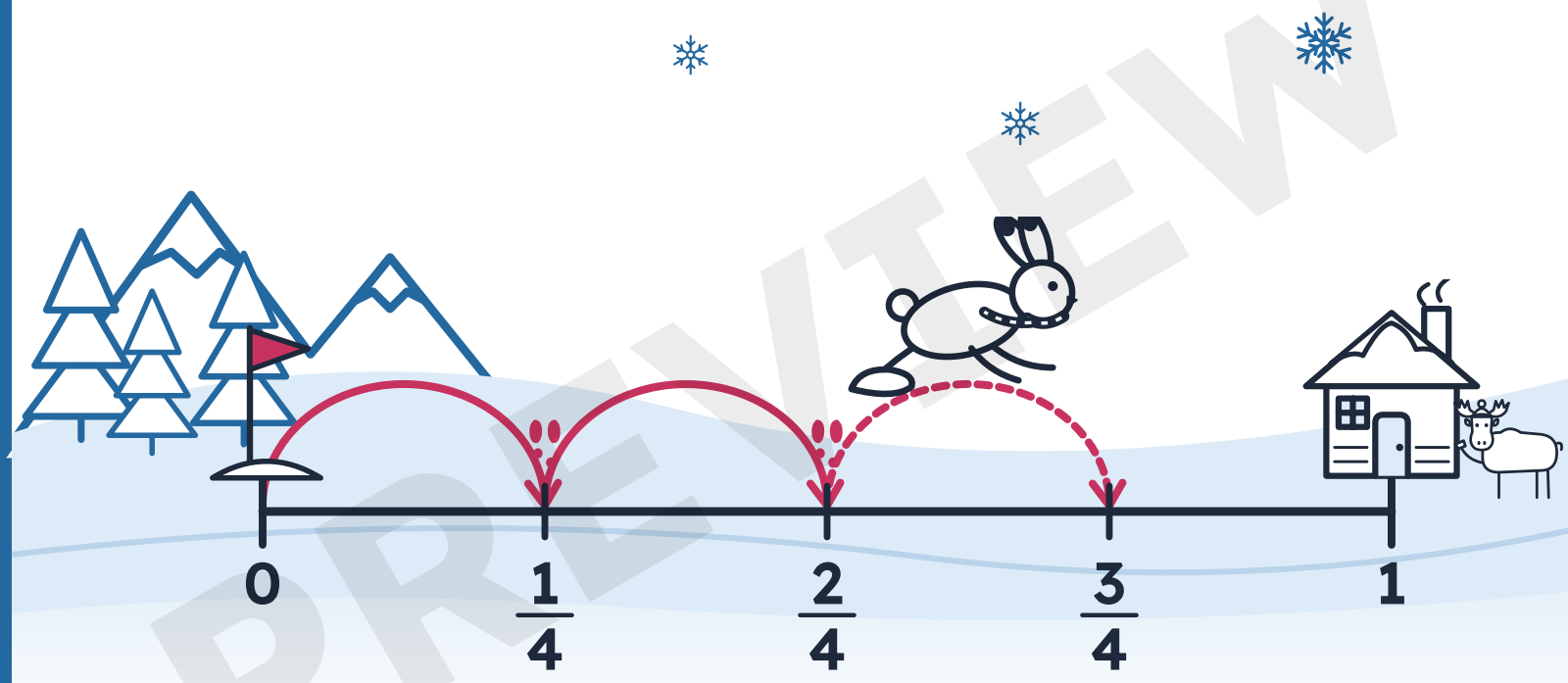
3rd Grade | 3.NF.A.2 & 3.NF.A.3

Fractions on a Number Line

Winter Trail Mazes & Practice



PREVIEW

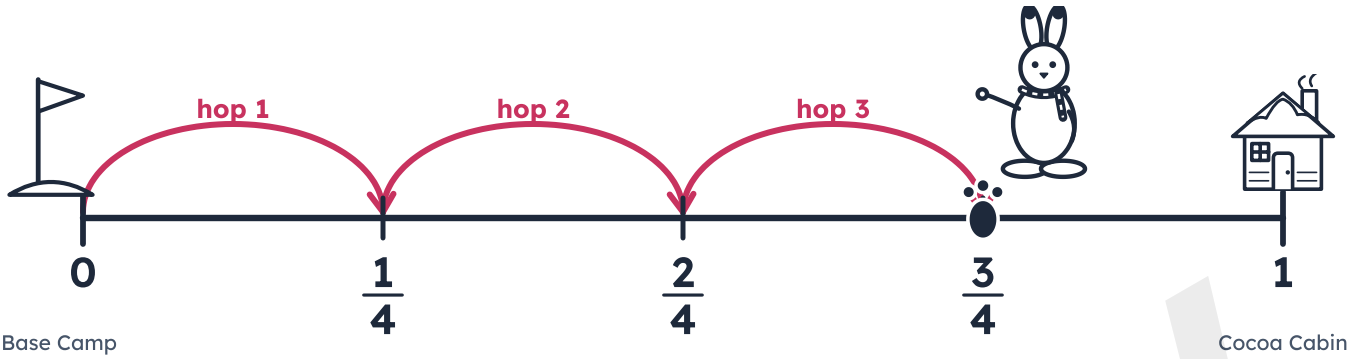
Every hop is the same size!

- 3 self-checking trail mazes
- Snowball Four-in-a-Row game
- Equivalent fractions & whole numbers
- Exit tickets + quick check
- Answer keys included

11 Student Pages + Anchor Chart | No Prep | Prints in Black & White

Trail Guide: Fractions on a Number Line

Count the hops. Name one hop. Hop to the spot.



1 Count the hops.

→ the denominator

There are 4 equal hops (equal parts) from 0 to 1. The **denominator** is 4.

2 Name one hop.

→ the unit fraction

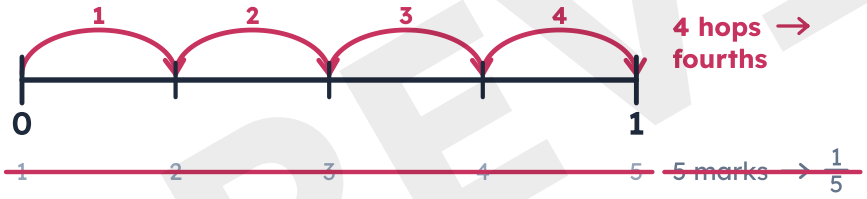
Each hop is $\frac{1}{4}$ of the whole.

$\frac{1}{4}$ is a **unit fraction.**

3 Hop to the spot.

→ the numerator

Start at 0. Juniper took 3 hops of $\frac{1}{4}$. The **numerator** is 3. She is at $\frac{3}{4}$.



Count the hops, not the marks!

0 is where Juniper starts. She has not hopped yet. 4 hops make 5 marks. Count the hops!



You reached the cabin!

4 hops of $\frac{1}{4}$ reach the cabin. $\frac{4}{4}$ = 1 whole.

Word Bank

number line	A line that shows numbers in order.
fraction	A number that names equal parts of a whole.
numerator	The top number. It counts hops from 0.
denominator	The bottom number. Equal hops from 0 to 1.
equal parts	Parts that are all the same size.
unit fraction	One equal part, like $\frac{1}{4}$.
whole	The whole trail from 0 to 1.
tick mark	A short line that marks a spot.
interval	The space between two tick marks. One hop!
equivalent	Equal. Worth the same amount.
same point	One spot on the line with two names.

PREVIEW

Name: _____ Date: _____

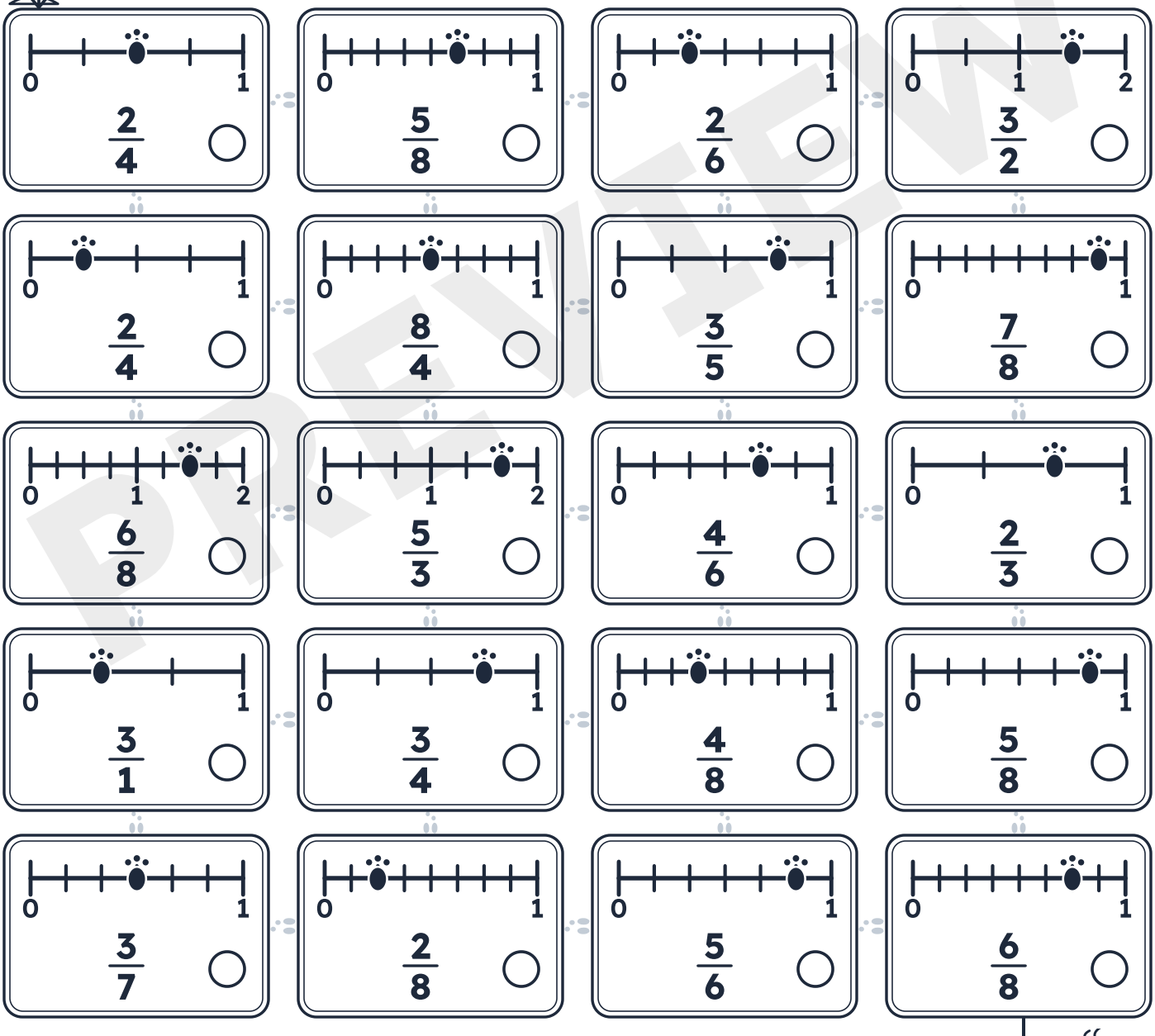
Trail Maze 2: Fractions on a Number Line

Only true trail signs lead home.

Directions: Start at Base Camp. Read each trail sign. Does the fraction name the footprint? If YES, color the circle. Hop to the next sign: up, down, left, or right. No diagonal hops. Only true signs lead to the Cocoa Cabin!

Some trails go past the cabin to 2. The whole is still 0 to 1.

Base Camp
START here



Self-check: The right trail has 12 footprints.
Count yours! I colored _____ footprints.

Cocoa Cabin
FINISH

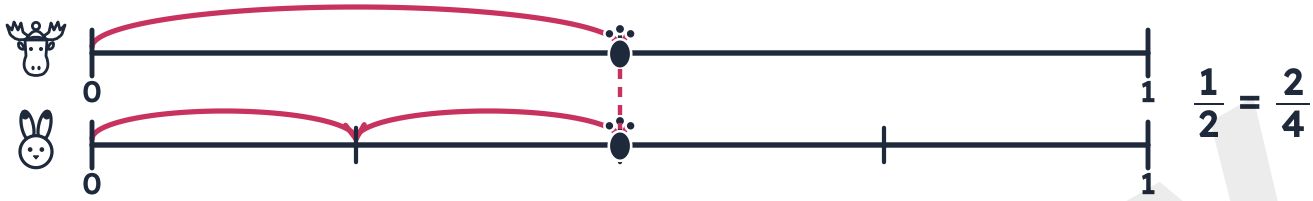
PREVIEW

Name: _____ Date: _____

Same Spot, Same Number: Equivalent Fractions

Birch takes one big step. Juniper takes two hops.

Part A: Same Spot. Find where Birch is. Draw Juniper's footprint at the same spot. Fill in the fraction.



$\frac{1}{2} = \frac{2}{4}$

Example: Birch took 1 step of $\frac{1}{2}$. Juniper took 2 hops of $\frac{1}{4}$. They land on the same spot, so $\frac{1}{2} = \frac{2}{4}$.

1. Birch is at $\frac{2}{3}$. Draw Juniper's footprint at the same spot.



$\frac{2}{3} = \frac{\square}{6}$

2. Birch is at $\frac{3}{4}$. Draw Juniper's footprint at the same spot.

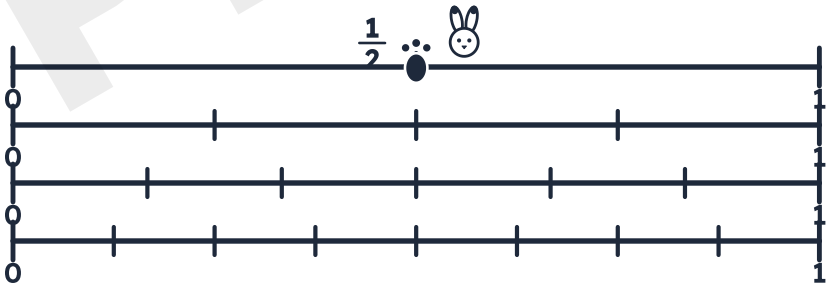


$\frac{3}{4} = \frac{\square}{8}$

They land on the same spot, so _____

Part B: Many Answers. Use the trails. Find fractions at the same spot.

3.



Juniper is at $\frac{1}{2}$. Find 3 more fractions that land on the same spot.

$\frac{1}{2} = \frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{\square}$

I know they are equal because _____

Part C: Reach the Cabin. Write fractions that name a whole number.

4. Write 3 different fractions that land on the Cabin (1 whole).

$\frac{\square}{\square}$ $\frac{\square}{\square}$ $\frac{\square}{\square}$



5. Birch is at Cabin 2. Write 2 fractions that name his spot.

$\frac{\square}{\square}$ $\frac{\square}{\square}$



PREVIEW

Name: _____ Date: _____

Fix the Trail Sign: Find the Mistake

Someone mixed up the signs!

Directions: Read each trail sign. Circle **Correct** or **Needs a fix**. Tell why. Then draw the right trail.

Nadia's trail sign



Nadia's sign says: **"The footprint is at $\frac{1}{5}$."**

Circle one: Correct Needs a fix

The mistake is _____

_____

Draw it right:



Emeka's trail sign



Emeka's sign says: **"The footprint is at $\frac{1}{3}$."**

Circle one: Correct Needs a fix

The mistake is _____

_____

Draw it right:



Lucia's trail sign



Lucia's sign says: **"The footprint is at $\frac{3}{6}$."**

Circle one: Correct Needs a fix

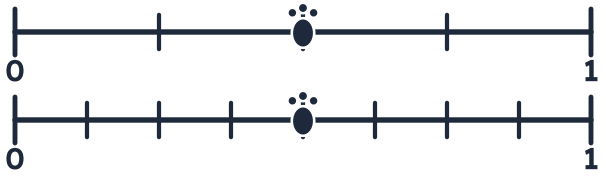
The mistake is _____

_____

Draw it right:



Tobias's trail sign



Tobias's sign says: "$\frac{2}{4} = \frac{4}{8}$ **because they are the same point.**"

Circle one: Correct Needs a fix

I know because _____

_____

Show why:



PREVIEW

Team Snowball: _____ Team Mitten: _____

Snowball Four-in-a-Row: Fractions

Name the point. Claim the square! (2 players)

Snowball = O

Mitten = X

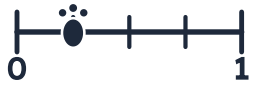
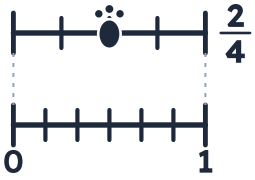
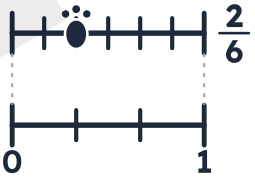
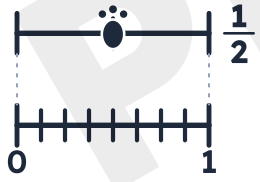
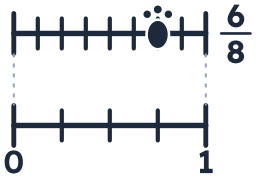
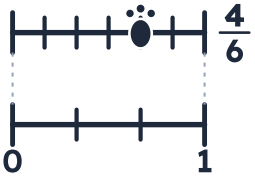
How to play

1. Pick an open square. Say the fraction.
2. Your partner checks the Answer Card.
3. Correct? Draw your mark (Team Snowball: O, Team Mitten: X).
4. Get 4 in a row across, down, or diagonal to win!

F Footprint: Name the footprint. Use the trail's hops.

T Two trails: Name the bottom fraction at the same spot.

C Cabin: The footprint is on a whole number. Name it as a fraction. Use the trail's hops.

F 	T 	F 	C 	F 
C 	F 	F 	T 	F 
T 	F 	C 	F 	F 
F 	F 	T 	F 	C 
F 	C 	F 	F 	T 

PREVIEW